MATTHEW H. ZHANG

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EDUCATION

The University of Texas at Austin – GPA: 3.52 May 2029
Bachelor of Science in Electrical and Computer Engineering Honors
Bachelor of Business Administration in Canfield Business Honors
Relevant Coursework: Circuit Analysis & Design Honors, Computing Systems Honors, Embedded Systems Honors, Software Design & Implementation Honors, Data Science Honors, Management Information Systems Honors

EXPERIENCE

Member of Technical Staff – *Stealth Robotics Startup ($25M+ raised, Kleiner Perkins)* Jun 2026 – Present

- Led tactile-sensing R&D across magnetic, optical, capacitive, and resistive approaches; built piezoresistive thru/shunt-mode FPC (flex PCB) prototypes and a calibration rig from a repurposed 3D printer, load cell, and fiducial vision tracking
- Owned schematic, layout, firmware, and bring-up of a 40x40mm STM32 CAN FD-UART bridge with 24V-to-11V 6A buck
- Schematic capture of 1 kW 24V/50A bimanual PDB + USB-hub with triple buck rails and 5 Gbps data transmission
- Deployed CV gripper calibration tools, reducing operator downtime 92% and increasing per-station data capture 110%
- As the only electrical engineer, assisted mechanical development and fabrication of 150+ data-collection devices

Co-Founder – *Inhabit, NVIDIA Inception Startup & Founders Inc Residency* Dec 2025 – Jun 2026

- Led development of a wearable motion capture system streaming human joint angles at 50 Hz over USB serial
- Designed MCU-based magnetic encoder board using RS-485 daisy-chaining, enabling plug-and-play joint modules
- Deployed and validated on Unitree G1 EDU for episodic task collection; redesigning V2 for robustness and UX

Lead Hardware Engineer – *Texas Aerial Robotics, Undergraduate Research* Jan 2026 – Present

- Modeled inverted pendulum and TVC system dynamics and implemented PID, MPC, & LQR controllers in Python
- Designing compact AM32-based 30A 4S BLDC motor driver PCB under size and thermal constraints

Embedded Systems Engineer – *GHOST Robotics, VEX U* Nov 2025 – May 2026

- Redesigned differential I2C sensor bridge PCB with address translation and reduced board area by 40%
- Wrote interrupt-based RP2040 sensor polling firmware in C with custom sensor drivers over USB serial
- Performed board bring-up and debug by validating power rails and signals with a multimeter and logic analyzer

Team Captain & Engineering Lead – *BagelBytes Robotics FRC #9400* May 2024 – Aug 2025

- Mentored 20+ members in CAD & electrical systems; improved team ranking by 1,500+ places during leadership
- Designed 100+ custom sheet metal parts, 7 custom-built gearboxes, four-bar linkages, cascade elevators, and more
- Engineered custom CAN bus harness and architecture, reducing match-time electrical failures by 28%

Venture Analyst Intern – *Tess Ventures* Mar 2026 – Jun 2026

- Evaluated robotics startups and provided technical due diligence for investment decisions
- Prepared technical demos and product-facing presentations for engineering and business audiences

PROJECTS

BetaGo, Autonomous Chess Robot – 1st Place @ World Tour Physical AI Hackathon May 2026

- Implemented classical, VLM, VLA, and IK control methods to autonomously play chess with a robot arm
- Introduced CNN segmentation and end effector adjustments to increase pick and place accuracy by 61%

Vinyl Record Player – Personal Project (In Progress) Dec 2025 – Present

- Designed and bench-tested a 2-layer ESP32 motor-driver PCB with USB-C PD power negotiation for the turntable
- Built the mechanical drivetrain (coreless DC motor, valve-guide main bearing, platter mount) with hall-effect RPM sensing and closed-loop PID speed control

SKILLS

Technical Skills: Circuit Design, PCB Layout, Firmware Development, Control Systems, Robot Learning, CAD, Git, Debugging, Soldering | **Tools:** KiCad, Altium, LTspice, MuJoCo, Linux, Onshape, GDB, Oscilloscope, Logic Analyzer, Garage Tools, Excel | **Languages:** C, C++, Python, Assembly, R, SQL | **Embedded:** CAN, I2C, SPI, UART, ADC/DAC, BLE

AWARDS

UC Berkeley AI Hackathon Best Physical AI Hack (2026) • Founders Inc Physical AI Hackathon 1st Place (2026) • UT x CSBA x HBA Hackathon 1st Place Track (2026) • IEEE RAS Robotathon 1st Place, Software Design Award (2025) • FIRST Robotics Rising All Stars Award (2025) • World's Rookie Inspiration Award, FIRST Robotics Worlds Qualifier (2024)